

"Put your feet in open pit" - a WebXR Unity application for learning about the technological processes in the open pit mine

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ABSTRACT

In the mining industry, accessing the locations and observing processes for educational purposes can be problematic due to safety reasons or high cost of experiencing it in person. Immersive technologies, especially applied in an accessible and responsive design such as WebXR, have great potential to make such educational and training applications easy to access and effective for a wide audience. In this work-in-progress paper, we present the design process, the detailed production pipeline and the challenges we have met during the implementation of the WebXR interactive environment based on the 360 video of the mining technologies pipeline captured at the open pit mine.

Index Terms: WebXR—360 film—Mining industry—Design thinking; Responsive design—Educational experience

1 INTRODUCTION

WebXR [17] is an API allowing providing Virtual Reality (VR), Augmented Reality (AR) or Mixed Reality (MR) experiences (jointly referred to as XR) via web browser for all devices that support these modes. It is a promise of simplification and unification of cross-platform XR development and access. These features make WebXR recognized as "real revolution in multimedia content for education" [29], democratizing the access to immersive education.

While the idea of WebXR and the possibilities it brings in a matter of accessibility, especially in education, are undoubtedly extraordinary, its road towards fully delivering the aforementioned promise remains bumpy when it is about providing appropriate support for all devices and web browsers. A qualitative study was conducted with WebXR creators [37] to understand their experiences and 8 key challenges were reported as a result. Relatively steep learning curve and small community around this technology, performance limitations of browsers, different levels of support by browsers, necessity

to keep up with frequent updates and differences among different devices were mentioned among the limitations.

A systematic review of MR application for education was provided in the *MiReBooks* project [33]. For mining industry, both VR and AR applications are available. Early prototypes of virtual reality training applications for the mining industry were proposed in 2009 [34], but majority of them were produced in past several years. *METS VR* is a mining evacuation training simulator in virtual reality for underground mines [23], which provides a cost efficient and safe alternative to the current model of evacuation training. A similar system [27] was evaluated and the trainee perceptions of the realism, success and usefulness were measured. Another example is a VR training system prototype for drilling in underground mines [36]. VR can be used also for remote work - such operation system for a fully mechanised coal-mining face using real-time data and collaborative network technology was introduced [35]. There are also attempts to explore the usability of MR for mining industry workers, e.g. it was suggested that current and future personnel can model and make predictions using past, current and future data sets [32].

The 360 film and the application presented in this work in progress paper are the part of the bigger project *RawMaterials@Schools-ESEE* and they aim at using immersive technologies to prepare accessible and easy-to-use interactive educational experiences for students. In this article, we explore the potential of using a WebXR approach to make the experiences responsively designed and widely available.

2 WEBXR FOR EDUCATIONAL PURPOSES

WebXR has been recently successfully applied in biology and chemistry as the interactive model-viewer library. *MolecularWeb* [28,30] is a free, open-source website with interactive AR webpage-based apps and it allows students and teachers to naturally handle virtual objects to explore molecular structure, reactivity, dynamics, and interactions, covering topics from inorganic, organic, and biological chemistry. *BioSIM^{AR}* was developed as a free online software program to provide a new way to visualize and interact with 3D molecular models in a virtual environment using AR [24]. A new web portal, Cryo-Electron Tomography Online or *CETO*, was created to provide engaging online training and protocols for cryo-ET imaging which visualizes high-resolution three-dimensional structures while preserving the biological context of the sample. [26]. A collaborative WebXR app was designed also for medical learning [22]. COVID-19-related social distancing impacts not only education but also students' wellbeing, e.g. anxiety. Such problems are addressed by another WebXR application [25].

Apart from life sciences and social sciences applications, the great

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potential of WebXR has been noticed in urban planning and smart city management, e.g. [31]. The authors found out that XR environments that resemble common practices and situations encountered in everyday life of people participating in the planning process provide the most suitable framework for seamless integration, while environments with new digital affordances like AR measurements or semi-transparent walkable scale models were rated lower.

WebXR potential for spatial presentation can be explored also for geoengineering applications. WDR VR Bergwerk [5] is a WebVR project, a digital reconstruction of the last coal mine in the Rhine-Ruhr region, closed in 2018. Instead of guiding the user linearly through a story, as is the case with many other 360 video productions, at glueckauf.wdr.de the viewer can experience the mine interactively, individually and on their own initiative. Production of this type of experience was particularly hard since the use of electrically or electronically operated cameras underground is usually strictly prohibited because of the risk of a firedamp explosion. The project is supplemented by an interactive VR experience “Meet the Miner – WDR VR Bergwerk”, which is available on Steam and has met positive community reception [6].

3 PRODUCTION PROCESS

3.1 360 video

The first immersive product of our project is a 7-minute 360 video (English version available at [20]) recorded at Limestone Quarry Małogoszcz with Insta360 One X at the resolution of 5760x2880@30fps for the 360 view and with Insta360 One R at 3840x2160@30 for the first-person perspective parts.

The video shows the complete transport cycle in the mining plant realized by rigid haul truck. The video draws user’s attention to the important characteristic points of the cycle, such as:

- beginning of the haulage cycle;
- entry to the quarry (Fig. 1);
- arrival at the mining front;
- haul truck maneuver;
- haul truck loading (Fig. 2);
- returning with the material;
- material dumping to the hopper (Fig. 3);
- end of the haulage cycle.

Additionally, thanks to the inserted picture-in-picture films, other processes, necessary to be carried out before the transport process are shown during the video. These are:

- blastholes drilling;
- blasting works - detonation (Fig. 4);

The post-production, thanks to the Adobe Premiere Insta360 plugin doing proper video auto-stitching, focused on:

- providing optimal flow of showing technological processes;
- including their corresponding picture-in-picture videos enabling deeper insights into specific parts of the processes;
- manipulating the speed of some repetitive parts of the cycle or lowering the speed in order to show the details;
- setting process parts descriptions accordingly together with technical parameters (e.g. real time of each part of the cycle, number of repetitions, body payload or capacity of the loader).



Figure 1: Mining front.



Figure 2: Haul truck loading.

The picture-in-picture technique allowed showing at the same time different perspective (e.g. while loading the haul truck we can see both the perspective from the haul truck, from the wheel loader and the third person perspective from outside). Further improvements such as better video resolution, details and frame-rate with use of Topaz Video Enhance AI [15] are in progress.

The 360 video was a separate product but also it played a role of low-fidelity prototype for a final interactive virtual experience.

3.2 Design process

For designing the final set of functionalities to implement in WebXR we have conducted a design thinking process based on user-centered design paradigm. We have mixed several methods of prototyping, starting from physical prototypes (dioramas) in 3D (see Fig. 5) and supplementing it with 2D mockups to design the information architecture and points of interest (see Fig. 6). In the process we focused on the empathisation and ideation phases. The final concept assumed changing the passive linear video to an interactive journey on the 3D map, giving the user more affordances and possibility to explore on their own or follow the main scenario.



Figure 3: Material dumping to the hopper.



Figure 4: Blasting works - detonation.

3.3 WebXR application

For Unity terrain we used very high quality mesh based on geodata Digital Elevation Model heightmap (see Fig. 7), transformed into Unity terrain and optimized with use of Mesh to Terrain [7]. For the terrain shaders we selected the MicroSplat - Ultimate Bundle [8] paired with Better Lit Shader [1] for other models. For the textures we picked various substances from Adobe Substance 3D Assets [13] imported into Unity with use of Substance in Unity [14]. There are also some of the Quixel Megascans [9] in the phase of testing (if used, their meshes will most probably be optimized first with use of Simplygon [12]). Depending on the framerate of the final version of the prepared scenes on mobile devices, also Zibra Liquids real-time fluid simulation with use of AI-based object approximation [21] will be used for more realistic look of the scenes.

Since this is a work-in-progress, some of the aforementioned assets may be removed or replaced, depending on the performance tests on different devices. Future improvements of the described work will also include Unity Render Streaming [10]. The functionalities developed during the implementation include e.g.: two process exploration modes - first-person interactive mockup view and 1:1 scale first-person navigation through the map; quizzes with results saved to the cloud for optional further processing; 360 orbs with videos and photographs corresponding with each specific process phase.

The application is being made with use of LTS version of Unity 2020 under Universal Render Pipeline (URP) and for the Unity WebXR part we use the WebXR JavaScript API integration to Unity WebGL, specifically WebXR Export [18] and WebXR Interactions [19]. For various functionalities that do not work well straight out of the box for Unity WebGL third-party assets are currently being used: GyroscopeAccelerometerWebGL [4], Video Player WebGL [16], ScreenOrientationWebGL [11].

3.4 Challenges

Most of challenges regarding the current Unity WebXR approaches that we met are related to the use of the WebXR app with mobile devices. After the mobile VR passive HMDs got stopped being produced by their main manufacturers the current state of the mobile VR is that on one hand there are more and more powerful smartphones that are eventually quite "mobile VR ready" and on the other hand if someone wants to benefit from having smartphone for entry-level VR use - it is not easy (but possible) to get access to appropriate passive HMDs. If someone wants to keep the support for mobile VR as much as possible - this is where some challenges occur almost instantly during the WebXR Unity app development.

The accelerometer and gyroscope use - necessary for mobile to enable looking around in VR - needs to be implemented, it does not work by default. Access to the camera of the iPhone, if the app is designed to also use the AR mode, also requires some work. Getting the WebXR Unity app to go fullscreen is not easy, especially with mobile version of Safari (current state of mobile fullscreen support [2]). Also, when looking at the support for the WebXR



Figure 5: Low fidelity 3D prototyping - a physical prototype and a digital prototype for spatial design.



Figure 6: Low fidelity 2D prototyping - conceptual design.

itself [3], it is clear that WebXR needs some time for broader support, however with use of aforementioned third-party assets dedicated for mobile Unity WebGL and several tweaks in the html Unity WebGL template - from initial tests it occurred that all of the main functionalities required for mobile VR work as intended. Mobile VR interactions also have to be separately designed due to presumed lack of dedicated buttons and controllers. Thermal throttling and battery depletion also remain factors worth addressing when making design and implementation choices.

4 DISCUSSION AND FUTURE WORKS.

The reason for choosing the WebXR approach for making the app is to provide educational experiences to as wide variety of end-users devices as possible while keeping the ease of use and access to the interactive experience achieved by automatic and semi-automatic adjustments of the app depending on the specific device type advantages and limitations. This means the rendered graphics quality and interface mechanics will be adjusted according to what specific end-user device type allows. Specifically, benefiting the 3DoF from gyroscope and accelerometer on smartphones, and the 6DoF from the outside-in and the inside-out tracking of the VR headsets, handy input mapping across all types of VR controllers, while still providing standard keyboard and mouse input.

Future improvements of the interface may also put the hand tracking and voice user interfaces options under considerations. The OpenXR standard for the XR devices is expected to help a lot in a matter of supporting the wider and wider variety of XR inputs types.

We also plan to conduct UX and usability tests with different groups of users, including measurements of the educational content retention by pre-test and post-test and comparing it to the control groups.

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Figure 7: The terrain snapshot and the terrain mesh based on geodata Digital Elevation Model heightmap.

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